Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

This is a design calculation, not a thorough examination. A pinned lug between two cheek plates under a load along the lug: tension on the net section through the hole, bearing of the pin on the hole, shear tear-out beyond the hole, and the pin in double shear and in bending, each against an allowable stress; the largest utilisation governs. This sheet does not replace the thorough examination and inspection of lifting equipment by a competent person (LOLER reg 9) or the marking, test and certification that the regulations require. It is available only while the lifting feature is enabled; its issue needs a competent-person attestation and a second independent check. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation.

WHAT IT CALCULATES	RELATION
Tension across the net section beside the hole	$\sigma_t = P/((W - d_h)\cdot t)$
Bearing of the pin on the lug	$\sigma_{br} = P/(d_p\cdot t)$
Tear-out shear beyond the hole, on two planes	$\tau_{to} = P/(2\cdot a\cdot t)$
Pin shear, in double shear	$\tau_p = P/(2\cdot\pi\cdot d_p^2/4)$
Pin bending, from the lever arm across the lug, the gap and the shackle	$M = (P/2)\cdot(t/4 + g + t_s/2)$; $\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$
Utilisation of each mode, its stress over the allowable from the material and design category in force (the sheet adds no design factor of its own)	$u = \sigma/\sigma_{allow}$

Basis: Pinned connections[1](Ch. 3); Shear, bearing and bending stresses[2](Ch. 3).

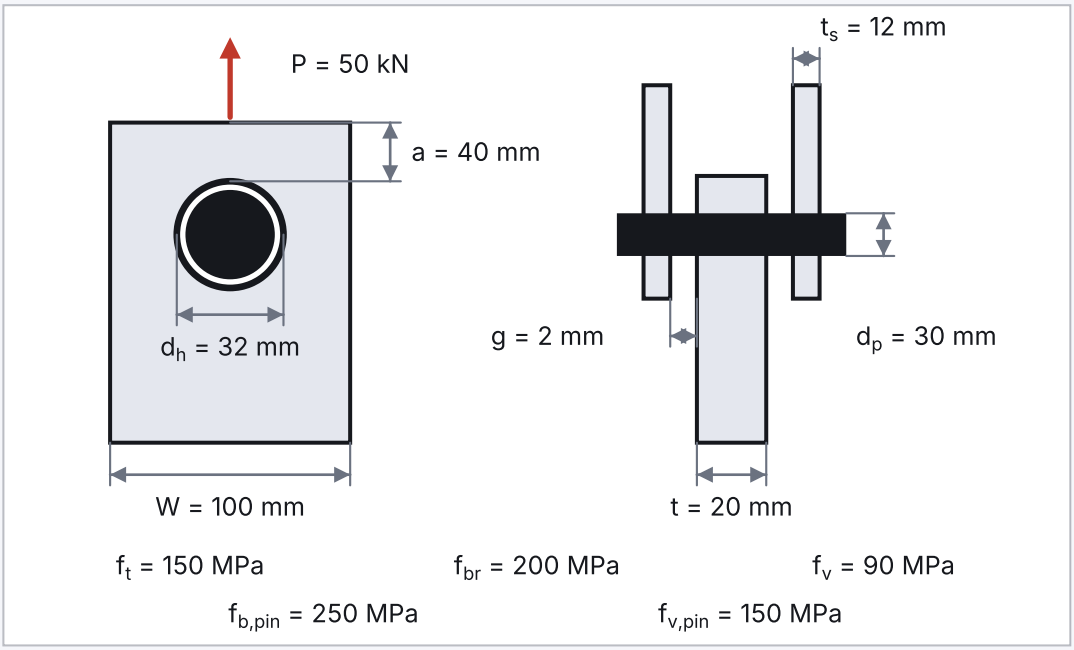


Figure 1 — A lug and its pin, front view and section through the pin: the lug between two cheek plates, the load through the pin. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Load on the lug (along the lug)	50	kN	spec: LOAD-SCHEDULE-01	0.01 … 100000 kN
t	Lug thickness	20	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
W	Lug width at the hole	100	mm	drawing: DRG-LL-101 rev A	5 … 2000 mm
d_h	Hole diameter	32	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
a	Edge distance, hole edge to lug end (load direction)	40	mm	drawing: DRG-LL-101 rev A	1 … 2000 mm
d_p	Pin diameter	30	mm	drawing: DRG-LL-101 rev A	2 … 1000 mm
t_s	Cheek plate thickness (each)	12	mm	drawing: DRG-LL-101 rev A	1 … 500 mm
g	Clearance, lug face to cheek plate (each side)	2	mm	drawing: DRG-LL-101 rev A	0 … 100 mm
f_t	Allowable tensile stress, lug	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
f_{br}	Allowable bearing stress, lug	200	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_v	Allowable shear stress, lug	90	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa
$f_{b,pin}$	Allowable bending stress, pin	250	MPa	spec: SPEC-MECH-03 design criteria	1 … 3000 MPa
$f_{v,pin}$	Allowable shear stress, pin	150	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate; each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A3. A close-fitting pin ($d_p \geq 0.9 \cdot d_h$). (Uniform-bearing idealisation.)
A4. A design calculation, not a thorough examination: it checks the lug and pin at the stated loads against the allowable stresses entered and nothing else. (LOLER reg 9 and PUWER duties are the owner's and the competent person's.)

4. Working

[14] $k_{fit} = d_p / d_h$
$= 30\ mm / 32\ mm$
$= 0.9375$
Pin-to-hole ratio

[15] $A_{net} = (W - d_h) * t$
$= (100\ mm - 32\ mm) \times 20\ mm$
$= 1360\ mm^2$
Net section through the hole

[16] $\sigma_t = P / A_{net}$ [1](Ch. 3)
$= 50\ kN / 1360\ mm^2$
$= 36.765\ MPa$
Net-section tension

[17] $A_{br} = d_p * t$
$= 30\ mm \times 20\ mm$
$= 600\ mm^2$
Bearing area

[18] $\sigma_{br} = P / A_{br}$
$= 50\ kN / 600\ mm^2$
$= 83.333\ MPa$
Bearing of the pin on the hole

[19] $A_{to} = 2 * a * t$
$= 2 \times 40\ mm \times 20\ mm$
$= 1600\ mm^2$
Tear-out area (two planes beyond the hole)

[20] $\tau_{to} = P / A_{to}$
$= 50\ kN / 1600\ mm^2$
$= 31.25\ MPa$
Tear-out shear

[21] $A_p = 2 * pi * d_p^2 / 4$
$= 2 \times \pi \times (30\ mm)^2 / 4$
$= 1413.7\ mm^2$

Pin area in double shear

$$[22]\ \tau_p = P / A_p$$
$$= 50\ \text{kN} / 1413.7\ \text{mm}^2$$
$$= 35.368\ \text{MPa}$$

Pin shear

$$[23]\ M_p = P / 2 * (t / 4 + g + t_s / 2)$$
$$= 50\ \text{kN} / 2 \times (20\ \text{mm} / 4 + 2\ \text{mm} + 12\ \text{mm} / 2)$$
$$= 325\ \text{N·m}$$

Pin bending moment (uniform bearing on lug and cheek plates)

$$[24]\ Z_p = \text{pi} * d_p\text{^}3 / 32$$
$$= \pi \times (30\ \text{mm})\text{^}3 / 32$$
$$= 2650.7\ \text{mm}^3$$

Pin section modulus

$$[25]\ \sigma_b = M_p / Z_p$$
$$= 325\ \text{N·m} / 2650.7\ \text{mm}^3$$
$$= 122.61\ \text{MPa}$$

Pin bending stress

$$[26]\ u_t = \sigma_t / f_t$$
$$= 36.765\ \text{MPa} / 150\ \text{MPa}$$
$$= 0.2451$$

Net-section utilisation

$$[27]\ u_{br} = \sigma_{br} / f_{br}$$
$$= 83.333\ \text{MPa} / 200\ \text{MPa}$$
$$= 0.41667$$

Bearing utilisation

$$[28]\ u_{to} = \tau_{to} / f_v$$
$$= 31.25\ \text{MPa} / 90\ \text{MPa}$$
$$= 0.34722$$

Tear-out utilisation

$$[29]\ u_p = \tau_p / f_{v,pin}$$
$$= 35.368\ \text{MPa} / 150\ \text{MPa}$$
$$= 0.23579$$

Pin shear utilisation

$$[30]\ u_b = \sigma_b / f_{b,pin}$$
$$= 122.61\ \text{MPa} / 250\ \text{MPa}$$
$$= 0.49043$$

Pin bending utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
σ_t	36.765 MPa	Net-section tension
σ_{br}	83.333 MPa	Bearing stress
τ_{to}	31.25 MPa	Tear-out shear
τ_p	35.368 MPa	Pin shear
M_p	325 N·m	Pin bending moment
σ_b	122.61 MPa	Pin bending stress
u_t	0.2451	Net-section utilisation
u_{br}	0.41667	Bearing utilisation
u_{to}	0.34722	Tear-out utilisation

SYMBOL	VALUE	DESCRIPTION
u_p	0.23579	Pin shear utilisation
u_b	0.49043	Pin bending utilisation
$sheet_{notice}$	"This is a design calculation, not a thorough examination. It does not replace the thorough examination by a competent person (LOLER reg 9) or the marking, testing and certification that the regulations require. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation."	Status of this sheet

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pin bending	u_b = 0.49043	$u_b \le 1$	0.49	
PASS	Bearing	u_{br} = 0.41667	$u_{br} \le 1$	0.417	[1](Ch. 3)
PASS	Tear-out	u_{to} = 0.34722	$u_{to} \le 1$	0.347	[1](Ch. 3)
PASS	Net-section tension	u_t = 0.2451	$u_t \le 1$	0.245	[1](Ch. 3)
PASS	Pin double shear	u_p = 0.23579	$u_p \le 1$	0.236	

7. Applicability

No warnings.

8. Verification

Template mech.lifting-lug@1.1.0 Content hash 919ed87aaa88e2d9 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.